

Conclusion: Evidence Landscape,
Methodological Limitations, and Research
Agenda

Summary

This work demonstrates a first-generation prototype infrastructure for computational meta-analysis of a rapidly growing scientific field. By combining multi-source retrieval ($N = 849$ papers from three databases), LLM-based assertion extraction encoded as nanopublications, and citation-weighted hypothesis scoring, we produce a queryable, RDF-compatible knowledge graph that tracks the evolving evidence for eight core Active Inference claims. The system demonstrates the feasibility of automated living reviews, while clearly delineating the boundaries of current model capabilities.

Constraints and Methodological Scope

Several conscious design constraints scope these findings.

Keyword Classifier Resolution

The keyword-based classifier operates over 65+ mathematical indicators distributed across 8 domain categories, using a deterministic priority system that routes papers to specific application domains (C1–C5) before testing tools (B), formal theory (A1), and the qualitative philosophy catch-all (A2).

Word-boundary-aware matching reduces partial-match false positives, but keyword-based methods cannot capture semantic nuance: papers using novel terminology or discussing cross-domain topics without standard vocabulary risk misclassification. Residual A2 concentration should be interpreted as a ceiling on broad theoretical generality rather than a literal measure of philosophical focus. An embedding-based classifier trained on a labeled subset would provide a quantitative upper bound on the fraction of A2 papers that merit redistribution.

Citation Network Coverage Gaps

Future Directions: Beyond Tally-Based Evidence Aggregation

The current scoring formula (described in the methodology) aggregates LLM-extracted assertions through a simple citation-weighted tally. While this approach provides a transparent and reproducible baseline, it leaves substantial room for methodological sophistication. We identify six directions, ordered by expected impact, with the first three specifically addressing the limitations of tally-based evidence synthesis.

Hierarchical Bayesian Hypothesis Scoring

The most direct extension replaces the additive tally with a **hierarchical Bayesian model** that treats each hypothesis score as a latent variable inferred from noisy assertion observations. Under this formulation, each assertion a_i contributes a likelihood term $P(a_i|\theta_H, \sigma)$ parameterized by the hypothesis-level evidence strength θ_H and an observation noise term σ capturing LLM extraction uncertainty. A hierarchical prior $\theta_H \sim \mathcal{N}(\mu_{\text{field}}, \tau^2)$ pools information across hypotheses, enabling principled shrinkage for hypotheses with limited evidence (e.g., 1/6 Clinical Utility, 1/6

Limitations

This meta-analysis is subject to three primary limitations that govern the interpretation of its findings. First, as a deliberate design trade-off for corpus-scale processing, we are only analyzing abstracts for claims. This abstract-only extraction inherently misses granular empirical evidence, methodological caveats, and nuanced findings embedded within full texts. Second, our analysis possesses limited bibliographic depth; keyword-based retrieval across disjoint APIs produces a snapshot that inevitably contains coverage gaps, cross-source identifier mismatches, and false negatives. Third, recognizing these constraints, the current findings do not constitute a definitive, static ledger of the field's evidence base. Rather, the work is presented as a living open-source milestone and a computational scaffold for ongoing collaboration. By publishing this architecture, we invite the community to iteratively curate custom bibliographies, refine extraction prompts, and gradually extend the system toward full-text analysis.

Broader Impact

The vision motivating this work is straightforward: a living literature review—a continuously updated knowledge graph tracking what a field claims, what evidence supports those claims, and where the frontiers of understanding lie. This vision builds on the foundation established by Knight et al. knight2022sep, who identified the development of systems that could “encompass increased scope of relevant works,” “integrate multiple forms of annotation and participation,” and “facilitate integration of manual and artificial contributions” as key goals for the field.

By demonstrating that LLM-driven assertion extraction can produce scalable, queryable representations of scientific evidence—processing $N = 849$ papers spanning approximately two and a half decades (2005–2026), extracting structured assertions, and evaluating 8 core hypotheses—this work provides a reusable architecture for realizing this vision. The corpus window begins in 2005 to capture Energy-Based Model and variational Bayesian antecedents that predate the Free Energy Principle label itself; the formal FEP was introduced in 2006 friston2006free and reached its